

The Width Barrier

A Conditional Decidability Result for $\mathcal{ALCI}_{\text{RCC5}}$, and the Unity of Its Two Open Directions

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Abstract

Concept satisfiability for $\mathcal{ALCI}_{\text{RCC5}}$ — the description logic $\mathcal{ALCI}$ over the five RCC5 base relations under abstract composition-table semantics — has been open since Wessel (2002/2003). This report records the current, honestly-delimited state of one line of attack. We establish a *conditional* decidability theorem: $\mathcal{ALCI}_{\text{RCC5}}$ satisfiability is decidable provided a single property holds — *bounded live width* (the obligation we call F6) — and we note that the soundness half of the underlying reduction is machine-checked in Lean 4 with no unproved steps. We then analyse F6 itself. Two facts about the RCC5 composition table, both exhaustively verified, drive the analysis: (i) the only compositions that *determine* a relation (are singletons) all involve a vertical (PP/PPI) argument, and PO is never a determined value; (ii) consequently the “determined” structure a presentation may carry for free (*shadows*) is confined to the vertical axis, while the costly *live* structure lives on the horizontal (DR/PO) axis. Building on these we show that the width barrier *reduces to a definability question*: F6 fails if and only if some $\mathcal{ALCI}_{\text{RCC5}}$ concept can *force* unboundedly large rigid horizontal configurations. The substrate for such configurations is free (every DR/PO network is realizable), and horizontal chains *block* exactly as vertical ones do; so the entire barrier concentrates into whether the logic can pin rigid horizontal coordinates. Finally we observe that this is precisely the obstruction on the *undecidability* side as well: the reason no grid-encoding transfers to the abstract semantics (Wessel 2002/2003; Lutz–Wolter 2006) is the same looseness. The decidability keystone and the undecidability obstruction are one fact seen from two sides. We do not resolve F6; we localize it, certify everything around it, and place it on the knife’s edge between the two outcomes.

1 Introduction

The Region Connection Calculus fragment RCC5 partitions the relationships between two non-empty spatial regions into five jointly-exhaustive, pairwise-disjoint base relations. The description logic $\mathcal{ALCI}_{\text{RCC5}}$ takes these five as its roles (inverse roles are absorbed, since each relation’s converse is again a base relation) and interprets them over *abstract* models: structures in which every pair of distinct elements is labelled by exactly one base relation, subject only to the RCC5 composition table and converse laws. Whether concept satisfiability in this logic is decidable has been open for over two decades [1, 2]. No undecidability reduction is known to apply to the abstract semantics, and no decision procedure has been certified.

This report is not a claimed resolution. It is a *structural status report*: an account of where the difficulty now sits, written to be handed to someone new. Its contributions are three.

1. A **conditional decidability theorem** (§5): $\mathcal{ALCI}_{\text{RCC5}}$ satisfiability is decidable if the *bounded live width* property F6 holds. The soundness direction of the reduction — that a finite, well-

formed, composition-closed, faithfully-labelled certificate yields a genuine RCC5 model of the input concept — is fully machine-checked in Lean 4.

2. A **reduction of F6 to a definability question** (§§6–7): using two exhaustively verified facts about the composition table, we show that F6 fails exactly when some concept can *force* unboundedly large rigid horizontal configurations, and that the combinatorial substrate for such configurations is otherwise free.
3. A **unification of the two open directions** (§8): the forcing question that F6 reduces to is the same obstruction that blocks every attempt to prove the problem *undecidable*. Bounding it one way yields decidability; exploiting it the other way would yield undecidability; and the reason neither has happened is a single fact about the composition table.

Throughout, we distinguish clearly between what is machine-checked, what is exhaustively verified by finite computation, and what is argued.

2 Preliminaries

2.1 RCC5 and its composition table

The five base relations are DR (discrete — disjoint interiors), PO (partial overlap — share a part, neither contained in the other), PP (proper part), PPI (inverse proper part), and EQ (equal). Converse is the involution $\overline{DR} = DR$, $\overline{PO} = PO$, $\overline{EQ} = EQ$, $\overline{PP} = PPI$, $\overline{PPI} = PP$. The composition $\text{comp}(R, S)$ collects every base relation that can hold between x and z given $x R y$ and $y S z$; it is derivable from the subset semantics on non-empty regions and is displayed in Table 1.

comp	EQ	DR	PO	PP	PPI
EQ	{EQ}	{DR}	{PO}	{PP}	{PPI}
DR	{DR}	{EQ, DR, PO, PP, PPI}	{DR, PO, PP}	{DR, PO, PP}	{DR}
PO	{PO}	{DR, PO, PPI}	{EQ, DR, PO, PP, PPI}	{PO, PP}	{DR, PO, PPI}
PP	{PP}	{DR}	{DR, PO, PP}	{PP}	{EQ, DR, PO, PP, PPI}
PPI	{PPI}	{DR, PO, PPI}	{PO, PPI}	{EQ, PO, PP, PPI}	{PPI}

Table 1: The RCC5 composition table, $\text{comp}(\text{row}, \text{col})$. Rows are the first argument, columns the second. Computed by exhaustive enumeration over finite region domains.

An *atomic RCC5 network* on a set V assigns to each ordered pair of distinct elements exactly one base relation $N(u, v)$, satisfying converse coherence $N(v, u) = \overline{N(u, v)}$ (R2) and composition closure $N(u, w) \in \text{comp}(N(u, v), N(v, w))$ for all distinct u, v, w (R3); under *strong-EQ* semantics EQ occurs only on the diagonal (R1). We use the standard external input of the whole subject:

Proposition 1 (Patchwork property; Renz–Nebel). *A path-consistent atomic RCC5 constraint network is globally realizable: it has a model.*

2.2 $\mathcal{ALCI}_{\text{RCC5}}$ syntax and semantics

Concepts in negation normal form are generated by

$$C, D ::= \top \mid \perp \mid A \mid \neg A \mid C \sqcap D \mid C \sqcup D \mid \exists R.C \mid \forall R.C,$$

where A is a concept name and $R \in \{\text{DR}, \text{PO}, \text{PP}, \text{PPI}\}$ (the reflexive EQ is not used as a role between distinct elements). An interpretation $\mathcal{I} = (\Delta, \rho, \cdot^{\mathcal{I}})$ has a non-empty domain Δ , an atomic RCC5 network ρ on Δ , and concept-name extensions $A^{\mathcal{I}} \subseteq \Delta$. The role reading is $(\exists R.C)^{\mathcal{I}} = \{x \mid \exists y (\rho(x, y) = R \wedge y \in C^{\mathcal{I}})\}$ and dually for $\forall$; because ρ is atomic, the extension of role R is exactly the set of pairs labelled R . C_0 is *satisfiable* if $C_0^{\mathcal{I}} \neq \emptyset$ for some such $\mathcal{I}$. This is the abstract composition-table semantics of [1]; it is what makes the problem different from the topological logics of [2], whose undecidability does not transfer.

3 The split-forest presentation and the two axes

The line of attack organizes any model along two axes. The *vertical* axis is nesting: PP and PPI form a forest, since PP is a strict partial order ($\text{comp}(\text{PP}, \text{PP}) = \{\text{PP}\}$, so PP is transitive, and it is irreflexive under strong-EQ). The *horizontal* axis is DR and PO: the relations between elements that are not nested in one another. A model thus presents as a forest of nested regions with horizontal cross-links between incomparable ones; this is the *split-forest presentation*. The entire difficulty of the problem is captured by one asymmetry, which we now make precise.

4 Shadows and live width

The composition table forces some relations and leaves others free. The crucial instance is the pair

$$\text{comp}(\text{PP}, \text{DR}) = \{\text{DR}\}, \quad \text{comp}(\text{PPI}, \text{DR}) = \{\text{DR}, \text{PO}, \text{PPI}\}. \quad (1)$$

The first is a singleton: if $u \text{ PP } w$ and $w \text{ DR } v$ then $u \text{ DR } v$ is *forced* — discreteness propagates *into parts*. The second is not: discreteness does not propagate into wholes. This one-directional determinism is the source of both the method’s power and its difficulty.

Definition 2 (Shadow / determined edge). In an atomic RCC5 network N on V , an edge $\{u, v\}$ is *determined* (a *shadow*) if its label is fixed by path-consistency from N restricted to $V \setminus \{u, v\}$ together with the two elements’ other edges. A sufficient local condition is the existence of $w \in V$ with $|\text{comp}(N(u, w), N(w, v))| = 1$. An edge that is not determined is *live*.

A shadow costs nothing in a finite presentation: it is recoverable and need not be independently stored or checked. Hence the size that matters for a blueprint is not a node’s degree but its number of live neighbours.

Definition 3 (Live width; the property F6). The *live width* of a split-forest presentation is the supremum, over its interface clusters, of the number of live occurrences that must coexist in one cluster. Property **F6 (Bounded Live Width)**: there is a computable K such that every satisfiable concept C_0 has a model whose split-forest presentation has live width at most $K(|C_0|)$.

5 The conditional decidability theorem

The reduction built over rounds of this project, and its soundness core now certified in Lean 4, yields the following. We state it at the level of certificates; the formal development is `formal/Round19Transport.lean` (roughly 4,500 lines, zero `sorry`, no added axioms).

Theorem 4 (Conditional decidability). *If F6 holds, then $\mathcal{ALCT}_{\text{RCC5}}$ concept satisfiability is decidable.*

Sketch, with the certified content marked. A *certificate* is a finite *catalogue* (core network, attachment templates with a gluing normal form, and a template-indexed steering function) together with a plan; unfolding a plan produces a split-forest presentation. The development establishes, *in the Lean kernel*:

1. *Generation by construction.* For a structurally sound catalogue and a valid plan, the unfolded certificate is well-formed and *faithful* (its pattern values agree with the recorded interface rows) — `build_wellformed`, `build_faithful`.
2. *Catalogue soundness.* A finite composition check on the catalogue alone (the *S-condition*, independent of the plan length) plus faithfulness implies the full per-unfolding closure condition — `scat_scond` — hence the unfolded frame is a proper atomic RCC5 network (composition-closed, converse-coherent, EQ-free): `frame_closed`, `catalogue_soundness`.
3. *Truth.* A Hintikka labelling of the frame with the target concept at a root yields a genuine RCC5 model of that concept — `truth_lemma`, `sat_from_hintikka`; and the induced frame is verified to satisfy R1–R3 (`certInterp_rcc5`).
4. *Faithful abstraction.* Satisfiability and Hintikka-realizability coincide (`satisfiable_iff_hintikkaP`): a model always induces a Hintikka labelling, so the certificate loses nothing.

The decision procedure enumerates finite catalogues, checks the finite S-condition and a Hintikka labelling, and accepts iff a valid one carries C_0 at a root. Soundness (accept $\Rightarrow$ satisfiable) is (1)–(3), certified. Completeness (satisfiable $\Rightarrow$ accept) needs every satisfiable C_0 to admit a *finite* catalogue; this is the obligation `CompletenessObligation` in the development, and it follows from F6 (which bounds the catalogue, giving a finite search space and termination) together with a minor uniformization property $W2'$ affecting only extraction faithfulness — a failure of which can only cause over-rejection, never unsoundness. $\square$

5.1 Soundness and completeness: what is certified

It is worth separating the two directions of the reduction and stating exactly what the Lean development establishes for each, since the completeness side is more subtle than a single “machine-checked” would suggest.

Soundness — a valid finite certificate yields a genuine RCC5 model — is machine-checked in full (items (1)–(3) above: `catalogue_soundness`, `frame_closed`, `sat_from_hintikka`).

Completeness splits into two layers, and they have different status.

- **Abstraction completeness (infinitary) — certified.** Allowing an arbitrary, possibly infinite Hintikka labelling, completeness *is* machine-checked: `satisfiable_iff_hintikkaP` proves that C_0 is satisfiable *if and only if* some RCC5 frame carries a Hintikka labelling with C_0 at a point — both directions, no `sorry`. The non-trivial half, `model_hintikkaP` (every model induces a Hintikka labelling), is immediate once one observes that the Hintikka clauses are exactly the recursion clauses of satisfaction (§5, item (4)). Thus the type-system abstraction is *provably faithful*: it captures satisfiability exactly and loses nothing relative to models.
- **Finitary completeness — open, and equal to $F6 \wedge W2'$.** Requiring a *finite* certificate, as decidability demands, completeness is precisely the obligation `CompletenessObligation`: every satisfiable C_0 admits a finite catalogue, plan, and Hintikka labelling with C_0 at a root. This is *stated* in the development but deliberately *not proved* and *not axiomatized*; it is the conjunction of F6 (bounded width, which makes the catalogue finite and the search terminating) with $W2'$ (uniformization, which makes finite extraction faithful, and whose failure can only over-reject).

Direction	Lean witness	Status
Soundness: certificate $\Rightarrow$ model	<code>sat_from_hintikka</code>	certified
Completeness (abstraction): model $\Leftrightarrow$ Hintikka	<code>satisfiable_iff_hintikkaP</code>	certified
Completeness (finitary): sat $\Rightarrow$ finite certificate	<code>CompletenessObligation</code>	open = F6 $\wedge$
Decision-grade reduction (shape only; premise oracle-inhabitable)	<code>BoundedDecider.decidable</code>	certified*

In one line: the reduction is *sound* and *complete modulo* $F6 \wedge W2'$ — with soundness and the infinitary abstraction-completeness both certified, and the sole unformalized remainder being exactly $F6 \wedge W2'$, which the rest of this report analyses.

Remark 5. Theorem 4 is a genuine reduction, not a restatement: F6 is a single, precisely-posed property, and everything around it is machine-checked. What is *not* established is F6 itself (with the minor $W2'$). The rest of this report analyses it.

Remark 6 (Fourteenth review, GPT-5.5 cold). The first cold review of the Lean development (July 2026) confirmed the *adequacy of the soundness pipeline* — correct NNF syntax, correct `sat` polarity and role reading, the correct strong-EQ frame `RCC5Interp`, and correct composition/converse tables, with no soundness defect and no vacuity. Its findings are formalization-level and lie on the completeness interface: the certificate structures carry higher-order *function* fields rather than finite syntax; `Satisfiable` fixes the carrier to the occurrence type rather than arbitrary domains; and `CompletenessObligation` is an unbounded existence rather than a bounded, decidable search statement. None is a second mathematical conjecture beyond $F6 \wedge W2'$; each is the *under-formalization* of the finitary step, and each is repairable (finite codes with total evaluators; a countable-model bridge; a decision-grade bounded obligation with an executable checker). The review also confirmed one concrete over-restriction (the catalogue check inspects unused diagonal template values; machine-verified, completeness-side only). Thus the “machine-checked surround” of Theorem 4 is *exact* for soundness and the infinitary abstraction-completeness, while the finitary reduction still requires the finite-code restatement above — work orthogonal to, and downstream of, F6.

Remark 7 (Decision-grade reduction, now formalized). Responding to that review’s central point, the decision-grade reduction is now kernel-checked in the artifact. A `BoundedDecider` bundles a computable bound, a decidable Boolean checker, and the two correctness laws — a passing code witnesses satisfiability, and every satisfiable concept has a passing code below the bound — and `BoundedDecider.decidable` *derives* `Decidable` (`Satisfiable C0`) by a finite search over the range below the bound. The conditional theorem is thereby genuinely decision-grade: `Nonempty BoundedDecider  $\Rightarrow$  Decidable`. What remains is to *inhabit* `BoundedDecider` — precisely F6 (a computable width bound making the search finite), $W2'$ (faithful finite extraction), and the finite-coding of certificates — but the passage from a bounded decidable certificate to actual decidability is no longer a gap. (This is the reduction the fourteenth reviewer’s own sketch gestured at but could not kernel-check for want of a Lean installation.)

Independently, the review’s carrier finding (F2) is now closed: `Interp` and `Satisfiable` are carrier-polymorphic, so `Satisfiable` is *arbitrary-domain* (reference) satisfiability — $\exists \alpha \exists I : \text{Interp } \alpha \dots$ — with the occurrence-based pipeline as the $\alpha = \text{Occ}$ instance. The decision-grade theorem therefore concerns reference satisfiability, and the carrier bridge (reducing an arbitrary-domain model to a certificate) joins F6, $W2'$ and finite-coding on the single open completeness side.

Remark 8 (Finite-coding, now formalized (F1)). The review’s remaining structural finding (F1) — that `Cert`, `Template` and `Catalog` carry higher-order *function* fields rather than finite syntax — is now addressed in the artifact. The general fact is proved that a function is *finite data on a finite domain* (`fn_finitely_coded`: every f has a finite table reproducing it exactly on any finite set), and the finite code types (`FinTemplate`, `FinCatalog`, all first-order data) with *total* decoders are supplied, exactly as the review recommended. Faithfulness is proved for the first-order fields — `template_net_coded` and `coreNet_coded` reconstruct `net` and `coreNet` on their inspected port/index domains — and the higher-order steering field is handled by `f_factors_through_rows`: under the round-20 `f_reads_rows` clause, f depends only on the *finite* row-restrictions, so it too is finitely codable. (It is a pleasing closure that the clause the soundness proof was forced to add in round 20 is exactly what makes the steering function finite data.) What remains is a finite-code bridge into the pipeline (see the fifteenth-review correction below).

Remark 9 (Fifteenth review, GPT-5.5 cold — corrections). A cold review of rounds 26–28 (‘papers/gpt5.5_round28_review/’) returned *gap*, *repairable but not cosmetic*, and it is correct; two overclaims above are hereby withdrawn. **F2 is genuinely closed** (round 27). **F3 is only partially reduced**: the reduction `BoundedDecider.decidable` is correct, but its premise `Nonempty BoundedDecider` is inhabitable by a *classical oracle* (`check C _ := if Satisfiable C then true else false`, verified to compile), so it proves decidability from a classically vacuous premise rather than from a finite syntactic certificate scheme. **F1 is not closed**: round 28 supplies correct *local* tabulation lemmas and total decoders, but the normative completeness witness remains higher-order (including the labelling $\tau : \text{Occ} \rightarrow \text{List Concept}$), the finite codes are not wired into the checker, and there is no global inspected-domain coverage. The one *immediately* repairable part has been repaired: the non-vacuity witness (`certK_finitely_codable`), which the review correctly called *degenerate* (empty domain, zero templates), now proves *total* agreement — the decoded code reproduces `certK`’s core net on all indices and its entire two-template list — and a companion `fin_coding_nontrivial` exhibits the tabulation reconstructing a genuinely non-constant net (both constructive, no axioms). The substantive F1 gap (a finite-code bridge *into the checker*, and a finite-code τ) is untouched by this and remains open. Toward closing F3, the decidable heart of the checker is now in the artifact and machine-checked: `hintikkaB`, a Boolean Hintikka-checker over the finite frame, is proved *sound* (`hintikkaB_sound`: a passing check yields a genuine `Hintikka` system), and `sat_from_hintikkaB` strengthens the capstone so the labelling coherence — the liveness heart, the one acceptance test that is a finite computation yet had *no Decidable* instance as stated (the structural `Hintikka` quantifies over the infinite `Nat/Concept` types) — is discharged by that finite Boolean test rather than *assumed* (all axiom-clean: `propext/Quot.sound` only, no `sorry`, no choice). The residual, and it is the bulk of the finite-code bridge, is the finite-table decidability of `Wellformed` and `SCond` — equivalently `CatOk`’s `net_conv` and the core-net converse as finite-table checks, `AttachOk/PlanOk/SCat` as Boolean checks (each quantifies over finite lists), and the by-construction `F_reads` of a decoded `FinCatalog`. With those the fixed non-oracular checker becomes total and F3 closes; it is bounded, deliberate work, and honestly still open. **F4** also remains. Accordingly the claim “decidable modulo exactly $F6 \wedge W2'$ ” is *not yet honest*: closing the reduction additionally requires a non-oracular finite checker, a finite-code bridge for the certificate and the labelling, global domain coverage, and the F4 repair. What is genuinely established is narrower: soundness and infinitary abstraction-completeness (certified), the reference-semantics target (F2, certified), and a correct finite-search reduction *shape* awaiting an honest checker. The open mathematics F6 and $W2'$ are unchanged and remain the deep obstruction; the interface repairs above are finishable formalization, not new mathematics.

Remark 10 (Round-29: the non-oracular finite checker — F3 closed, F1 wired). The finishable

formalization above has now been finished, and machine-checked (all `propext/Quot.sound` only; no `sorry`, no choice; `native_decide` only in the witness). The artifact now carries a *fixed*, non-oracular Boolean checker `finAcceptB` of a *first-order finite* code — a coded catalogue (`FinCatalog`), a plan, a labelling *table*, and a root — that mentions neither `Satisfiable` nor any oracle, only finite table/decision checks and the Boolean Hintikka checker `hintikkaB`. Its soundness is a theorem (`finAccept_sound`: a passing check yields an RCC5 model, via the whole pipeline — converse tables give net-coherence, the bridge decoder gives `F_reads` by construction, the bounded fields are decided, `catalogue_soundness` and `build_wellformed` give `SCond/Wellformed`, and `sat_from_hintikkaB` gives the model). The decision-grade reduction `decidableSat_of_finScheme` then derives `Decidable` (`Satisfiable` C_0) from a fixed enumeration and a completeness premise *about the fixed checker*. Because the checker is fixed and forces exhibiting a real finite certificate, that premise is *not* oracle-inhabitable (a `Satisfiable` proof yields a possibly-infinite model, not a finite code) — unlike round-26’s free-field `BoundedDecider`, which it supersedes. Non-vacuity is witnessed (`finAccept_nonvacuous`: `finAcceptB` returns `true` on `certK` coded as a `FinCatalog`, with `atom 0` at the root). So the fifteenth review’s F3 (oracle-vacuity) is *closed*, and F1’s finite codes — catalogue *and* labelling — are now *wired into* a total checker. What remains is exactly the completeness premise itself: a *computable, complete* enumeration, which is precisely the open mathematics $F6 \wedge W2'$. This does *not* prove decidability; it makes “decidable modulo a complete finite-certificate enumeration” an honest, kernel-checked reduction, with the residual premise pinned to $F6 \wedge W2'$ (and F4, a completeness-side over-restriction, outstanding). Status unchanged: strongly supported, not certified.

6 The anatomy of forcing

We now isolate what the composition table can and cannot *force*. Call a composition $\text{comp}(R, S)$ *determining* if it is a singleton.

Proposition 11 (Determination is vertical). *Among the sixteen compositions of proper (non-EQ) relations, exactly four are determining:*

$$\text{comp}(\text{DR}, \text{PPI}) = \{\text{DR}\}, \quad \text{comp}(\text{PP}, \text{DR}) = \{\text{DR}\}, \quad \text{comp}(\text{PP}, \text{PP}) = \{\text{PP}\}, \quad \text{comp}(\text{PPI}, \text{PPI}) = \{\text{PPI}\}.$$

Every one of them has a vertical (PP or PPI) argument. No composition with both arguments horizontal ($\in \{\text{DR}, \text{PO}\}$) is determining, and PO is never the value of a determining composition.

Proof. Exhaustive inspection of Table 1; verified by finite computation. The horizontal block $\{\text{DR}, \text{PO}\} \times \{\text{DR}, \text{PO}\}$ gives $\text{comp}(\text{DR}, \text{DR})$ and $\text{comp}(\text{PO}, \text{PO})$ (all five relations) and $\text{comp}(\text{DR}, \text{PO}) = \text{comp}(\text{PO}, \text{DR})$ ’s companions of size three; none is a singleton. The four singletons are as listed, each with a PP/PPI leg. Scanning the table, no cell equals $\{\text{PO}\}$. $\square$

Proposition 11 explains a phenomenon observed experimentally: attempts to force a region to acquire unboundedly many neighbours succeed only in producing shadows. Any *forced* edge arises from a determining composition (or an intersection of constraints that pins a value), and by Proposition 11 the only fully determining compositions run through the vertical axis — so forced structure is shadow structure, confined to nesting. Concretely, a construction in which a distinguished “spy” region is forced discrete from every level of an ascending tower makes the spy’s degree grow without bound, yet each such edge is a shadow (forced by $\text{comp}(\text{PP}, \text{DR}) = \{\text{DR}\}$ through the tower); the *live* width stays bounded.

There is, however, a subtlety that keeps the barrier open. Determination and *distinctness* are different.

Proposition 12 (Distinctness can be horizontal). *Two same-typed occurrences are mergeable (identifiable, set to EQ) unless some constraint excludes EQ between them. A shared horizontal anchor excludes EQ: if $u \text{ DR } w$ and $w \text{ PO } v$ then u, v satisfy $\text{comp}(\text{DR}, \text{PO}) = \{\text{DR}, \text{PO}, \text{PP}\}$, which excludes EQ (forcing $u \neq v$) yet is not a singleton (the edge $\{u, v\}$ remains live).*

Proof. From Table 1, $\text{comp}(\text{DR}, \text{PO}) = \{\text{DR}, \text{PO}, \text{PP}\} \not\supseteq \text{EQ}$ and $|\{\text{DR}, \text{PO}, \text{PP}\}| = 3$. $\square$

Thus *determination* is strictly vertical (Prop. 11) but *distinctness* extends to the horizontal axis (Prop. 12). The two notions come apart, and it is precisely in the gap between them that F6 lives: a live horizontal crowd — many occurrences, pairwise distinct by horizontal anchors, with live (undetermined) mutual relations — is not ruled out by Proposition 11.

7 The width barrier reduces to horizontal forcing

We now follow the live-crowd possibility to its floor. Three facts, each finite-checkable, combine into a reduction.

Proposition 13 (The substrate is free). *Every $\{\text{DR}, \text{PO}\}$ -labelled network is path-consistent, hence realizable by Proposition 1. Moreover, for every n there is such a network on n nodes in which no two nodes are mergeable (all rows distinct): an irreducible live horizontal crowd of size n .*

Proof. For $R, S \in \{\text{DR}, \text{PO}\}$ one has $\{\text{DR}, \text{PO}\} \subseteq \text{comp}(R, S)$ (rows DR, PO of Table 1), so any $\{\text{DR}, \text{PO}\}$ -labelling meets every composition constraint: path-consistency is immediate. Realizability follows from Proposition 1. For irreducibility, a $\{\text{DR}, \text{PO}\}$ network is a graph (edge = PO, non-edge = DR); a graph with pairwise-distinct neighbourhood rows has no two mergeable nodes, and such graphs exist at every size. $\square$

Proposition 14 (Horizontal chains block). *The concept $\exists \text{PO}.\top \sqcap \forall \text{PO}.\exists \text{PO}.\top$, which demands an endless horizontal successor chain, is satisfiable by a finite model. More generally, any forced PO-chain admits a finite looping model.*

Proof. $\text{comp}(\text{PO}, \text{PO})$ is the whole algebra (Table 1), and in particular contains EQ; hence a PO-chain may close a loop ($w_i = w_j$ is composition-consistent). With finitely many concept types, any chain repeats a type, and a repeated type with mergeable surroundings closes the loop, exactly as in the vertical blocking (§3). A two- or three-element cyclic model satisfies the displayed concept. $\square$

Proposition 15 (No non-EQ horizontal recurrence — path form). *Form the composition-path automaton whose states are the reachable relation sets $\text{Rel}(w)$, with $\text{Rel}(\varepsilon) = \{\text{EQ}\}$ and $\text{Rel}(w \cdot a) = \bigcup_{r \in \text{Rel}(w)} \text{comp}(r, a)$, restricted to states not containing EQ. Every cycle in this restricted automaton uses only vertical labels (PP or PPI); no cycle uses a horizontal label (DR or PO).*

Proof. Exhaustive finite computation: there are 11 reachable relation sets, 8 without EQ, and the six cyclic edges among them are the self-loops $\{\text{PP}\}, \{\text{PO}, \text{PP}\}, \{\text{DR}, \text{PO}, \text{PP}\}$ under PP and $\{\text{PPI}\}, \{\text{DR}\}, \{\text{DR}, \text{PO}, \text{PPI}\}$ under PPI; none is horizontal. $\square$

Proposition 15 strengthens Proposition 11 from single compositions to whole paths, and sharpens Proposition 14: *any* support recurrence that never admits an equality fold is eventually vertical, hence absorbed by vertical cyclic unfolding, not by a horizontal crowd. It removes a class of would-be counterexamples but does not close F6, which is a statement about *width* rather than about paths. (This proposition was contributed by the fourteenth review and independently verified.)

Propositions 13 and 14 isolate the issue. A large live horizontal crowd is trivial to *exhibit* (Prop. 13) but cannot be produced merely by *demanding* more occurrences, because horizontal chains fold back on themselves (Prop. 14). To prevent the fold one would have to make the crowd members pairwise *non-mergeable in every model* — to give each a permanent address, a rigid horizontal coordinate. By Proposition 11 the composition table offers no lever to pin a horizontal coordinate: it cannot determine a horizontal relation, and PO is never forced. We arrive at the central structural statement.

Observation 16 (The width barrier is a forcing question). Bounded live width (F6) fails if and only if some $\mathcal{ALCI}_{\text{RCC5}}$ concept *forces* arbitrarily large *rigid* horizontal configurations — unboundedly many occurrences that must coexist at one interface and that no model may merge. Equivalently:

$$\begin{aligned} \text{F6 counterexample} &\iff \text{a concept forcing an unbounded rigid \{DR, PO\} graph,} \\ \text{F6 proof} &\iff \text{no concept forces such a graph.} \end{aligned}$$

The combinatorial substrate is free (Prop. 13); the mere generation of occurrences blocks (Prop. 14); so the whole barrier concentrates into whether the logic can force rigid horizontal coordination, a *definability* question about the expressive power of $\mathcal{ALCI}_{\text{RCC5}}$.

Observation 16 is a reduction, not a theorem in the formal sense: the forward direction (unbounded width $\Rightarrow$ forced irreducible crowds) is the definition of live width unwound; the content is that the converse obstruction is a pure forcing question, because everything *except* forcing — realizability and non-repetition — has been discharged by Propositions 13 and 14. We state it as an observation to mark that “forces” is used in its intuitive model-theoretic sense; making it a formal equivalence is itself part of the open problem.

8 One fact, two faces

The forcing question of Observation 16 is not new to the subject. It is the exact obstruction on the *undecidability* side.

To prove a logic undecidable one typically forces it to encode an unbounded, rigidly-coordinated grid on which a Turing computation (or a tiling) is written. For RCC8 with *topological* semantics this succeeds [2]. For the *abstract* composition-table semantics of $\mathcal{ALCI}_{\text{RCC5}}$ (and $\mathcal{ALCI}_{\text{RCC8}}$) it does not, and the reason was identified by [1]: the composition table cannot force the *coincidence condition* — that “east-of-north” and “north-of-east” name the same cell — because it cannot pin the relevant relations to single values. That is rigid horizontal coordination, and its unavailability is exactly Proposition 11: no horizontal determination, PO never forced.

Observation 17 (The knife’s edge). The bounded-width property F6 (decidability side) and the non-forcibility of a coordinate grid (undecidability side) are the same fact about $\mathcal{ALCI}_{\text{RCC5}}$: *the abstract composition table is too loose to pin rigid horizontal structure*. If this looseness can be turned into a bound, one obtains decidability (Theorem 4); if it could be circumvented and a grid forced, one would obtain undecidability. The two open directions are one question seen from two sides, and the reason neither has been settled in two decades is that the same fact must be turned one way or the other.

9 Discussion

Why the combinatorics did not fit. Attempts to attack the width bound with Ramsey-type combinatorics were unsuccessful, and Observation 16 explains why by type mismatch. Ramsey

theory concerns structure that is *unavoidable* in a large fixed object; F6 concerns structure that a formula can *force*. These differ in quantifier order: “every large graph contains X ” versus “some formula makes every model contain X .” The tools that fit the latter are model-theoretic: locality (Gaifman/Hanf), Ehrenfeucht–Fraïssé games, the composition method, and the finite/bounded-model-property machinery — together with the grid-encodability analysis in which [1, 2] already sit. This is a different toolbox from the model-*construction* techniques used to build the certificate layer, and it is the one Observation 17 indicates. We do not claim it succeeds; we claim it is the correct room to enter.

What is and is not established. Certified (Lean, no gaps): the soundness of the certificate-to-model pipeline and the faithfulness of the Hintikka abstraction (§5). Exhaustively verified (finite computation): Table 1 and Propositions 11, 12, 13, 14. Argued (structural, intuitive “forces”): Observations 16 and 17. Open: F6 itself, hence decidability. The honest one-line status is unchanged from the project’s standing label: *strongly supported, not certified*.

Value independent of F6. Theorem 4 is a conditional result that stands regardless of F6’s fate, with its soundness half machine-certified. Observation 17 is, to our knowledge, a new unification: it identifies the decidability keystone and the undecidability obstruction as one fact. Together they turn a formless two-decade question into a single, well-posed, two-sided lemma with a certified surrounding and a named toolbox. If the problem is ever settled we expect the resolution to begin from a map of this shape.

10 Conclusion

$\mathcal{ALCI}_{\text{RCC5}}$ satisfiability is decidable if the live width of its models is bounded (Theorem 4), and the soundness of that reduction is machine-checked. Whether the width is bounded reduces to whether the logic can force rigid horizontal coordination (Observation 16), which is exactly the obstruction that blocks proving the problem undecidable (Observation 17). The composition table’s inability to pin rigid horizontal structure is the single fact underneath both open directions. We have not turned that fact into a bound; we have shown that doing so, in either direction, is the whole of what remains.

References

- [1] M. Wessel. *On spatial reasoning with description logics — position paper* and technical report (report7), 2002/2003. Introduces $\mathcal{ALCI}_{\text{RCC}}$ under abstract composition-table semantics and identifies the coincidence-condition obstruction to grid encoding.
- [2] C. Lutz and F. Wolter. *Modal Logics of Topological Relations*. Logical Methods in Computer Science, 2006. Undecidability for topological RCC logics; the reductions do not transfer to the abstract semantics.
- [3] J. Renz and B. Nebel. *On the complexity of qualitative spatial reasoning: a maximal tractable fragment of the region connection calculus*. Artificial Intelligence, 1999. Source of the patchwork property.

Companion artifacts: the machine-checked development `formal/Round19Transport.lean`; the width-attack probes `verification/python/wp35_f6_width_attack.py`, `wp36_determination.vs.distinctness.py`, `wp37_f6_reduces_to_forcing.py`;

and the plain-language companion `WHY_ITS_HARD.md`. This report is deliberately at the boundary of formal and expository; where it and the machine-checked artifacts disagree, the artifacts win.